COURSE

IB Diploma Programme Business management SL

Investment appraisal

Unit 3: Finance and accounts · 3.8 Investment appraisal

Revision for Unit 3: Finance and accounts: each firm works it out on its own deals.

Class **IB Business Management SL** Topic **Investment appraisal**

THE LESSON

Setup				3 min
Round 1				Decisions 3 min · Deals 5 min · Topic tasks 4 min
Firm decisions	**Deals**	TASK 1 **Average profit**	TASK 2 **Limits of payback**	
Round 2				Decisions 3 min · Deals 5 min · Topic tasks 4 min · B2B deal 3 min
Firm decisions	**Deals**	TASK 3 **Rank by payback**	TASK 4 **Payback in months**	**B2B deal**
Round 3				Decisions 3 min · Deals 5 min · Topic tasks 4 min · B2B deal 3 min
Firm decisions	**Deals**	TASK 5 **Ramp or bay**	TASK 6 **Quicker payback**	**B2B deal**
Final questions				3 min

Fits a 30-minute lesson setting; longer settings add more deals.

Each step

Firm decisions	Each role makes its decision on its own page.
Deals	Each pupil trades with a classmate in another firm.
Topic task	The firm answers together. Everyone presses Agree.
B2B deal	One firm sells a lot of cars to another firm.
Final questions	Each pupil answers six questions.

Firm decisions

CEO	**Staff**	Hire staff. Each costs a salary and gives one benefit.
	Sabotage FROM ROUND 2	Hit a rival firm's cash, at a cost. A Legal Counsel blocks it.
Sales & Marketing	**Marketing**	Book one advert campaign to lift the next sales.
Finance	**Capital & Loans**	Borrow cash. The firm pays it back with interest.
	Investments FROM ROUND 2	Buy property or funds. The value can rise or fall.
Operations	**Seasonal stock**	Order the season's cars in advance.

B2B deal · Rounds 2 and 3

One firm sells a lot of 3 cars to another firm (4 in Round 3). With an odd number of firms, two firms offer to sell and the lower price wins.

Topic tasks

Each firm gets its own figures. The method is the same for every firm.

Task 1

ROUND 1

New charging points.

Average profit each year?

WORK OUT THE NUMBER · EXAMPLE FIGURES

Cost to set up	**£30,000**
Returns over its life	**£66,000**
It lasts	**4 years**

METHOD

Average profit = (returns – set-up cost) ÷ years

(£66,000 – £30,000) ÷ 4 = £9,000

Why: Average yearly profit is total profit shared over the years.

Task 2

ROUND 1

You choose between two projects by payback.

Which two are limits of payback?

PICK TWO

- ○ It favours slow projects
- ✓ **It ignores returns after payback**
- ○ It ignores the project's cost
- ✓ **It ignores the time value of money**
- ○ It is hard to work out

Why: Payback stops at the payback point and treats a pound later as a pound now.

Task 3

ROUND 2

Three projects with even yearly returns.

Order by payback, fastest first?

PUT THESE IN ORDER, FIRST TO LAST · RIGHT ORDER

1. **Cost £30,000, £12,000 a year**
2. **Cost £60,000, £20,000 a year**
3. **Cost £40,000, £10,000 a year**

Why: Payback: 2.5, 3 and 4 years.

Task 4

ROUND 2

A new valeting bay.

Payback period, in months?

WORK OUT THE NUMBER, THEN PICK IT · EXAMPLE FIGURES

Cost to set up	**£17,100**
Returns, each year	**£8,290**
Running costs, each year	**£1,450**

METHOD

Payback = cost ÷ (returns – running costs) × 12 months

£8,290 − £1,450 = £6,840; £17,100 ÷ £6,840 × 12 = 30

Why: It has paid for itself after 30 months.

Task 5

ROUND 3

Ramp: payback 2 years, ARR 10%. Bay: payback 4 years, ARR 18%. Demand is stable; cash is fine.

Which project should you choose, and why?

PICK A OR B AND GIVE A REASON

- ✔ **The bay**
- ○ The ramp

Why: With little risk and cash to spare, the higher return wins.

Task 6

ROUND 3

Which pays back first?

Payback of the quicker option, in months?

WORK OUT THE NUMBER · EXAMPLE FIGURES

Option A: cost	**£17,700**
Option A: returns a year	**£7,080**
Option B: cost	**£38,880**
Option B: returns a year	**£12,960**

METHOD

Each option: cost ÷ returns × 12 months; take the quicker

A 30 months; B 36 months; quicker 30

Why: The shorter payback gets the money back sooner.

Final questions

6
questions for each pupil

109
Level 3 Business questions

7
of them on investment appraisal

The game draws 6 of the 109 Level 3 questions; these 7 are about investment appraisal.

1 An inspection line costs £300,000 and should make £45,000 average annual profit. On original cost, what is its accounting rate of return?
15%.

2 A £180,000 paint booth returns £60,000, £75,000 and £90,000 over three years. Ignoring timing within years, what is its payback period?
2.5 years.

3 A project requires £120,000 now and has discounted future cash inflows of £138,500. What is its net present value?
£18,500.

4 A project costs £200,000 and its discounted future cash inflows total £224,000. What is NPV?
£24,000.

5 A system costs £250,000 and is expected to earn average annual profit of £30,000. What is ARR using original cost?
12%.

6 An energy-saving project costs £180,000. Discounted savings are £205,000, but a downside scenario reduces those savings by 15%. What is NPV in the downside scenario?
−£5,750.

7 A £90,000 paint booth brings in £30,000 in year 1, then £40,000 a year. What is its payback period?
2.5 years.